An Uncertainty-Based Outlier Detection Method for Satellite-Derived Land Surface Temperature Validation Using In Situ Measurements

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***Abstract*—Land surface temperature (LST) is a crucial parameter driving water and heat exchange at the surface–atmosphere interface. Satellite-derived LST requires rigorous validation to ensure its reliability in Earth system modeling and climate change research. To address validation accuracy degradation caused by cloud contamination artifacts and satellite–ground spatiotemporal mismatch errors, conventional mean- and median-based outlier detection methods were commonly used in the validation of satellite-derived LST products using in situ measurements. However, both methods are based solely on the degree of deviation within statistical data itself, without considering the uncertainties associated with satellite-derived and ground-based LST. This limitation could result in the biased identification of outliers in satellite-derived LST validation. In this study, an uncertainty-based method was proposed to detect outliers in the validation of Moderate Resolution Imaging Spectroradiometer (MODIS)-derived LST using in situ measurements. This method quantifies total LST uncertainty budgets to flag anomalous data points by integrating uncertainties from both satellite retrievals and ground observations. Validation results across surface radiation budget network (SURFRAD) sites demonstrate the method's efficacy when compared with those without outlier detection. Daytime implementation achieves significant root-mean-squared error (RMSE) reductions, notably at the Bondville (BND) site with a 3.1-K improvement, while nighttime applications yield marginal enhancements (<0.4 K), reflecting diminished thermal contrast and uncertainty components during nighttime. The uncertainty-based method consistently outperforms conventional mean- and median-based methods during daytime, with RMSE improvements ranging from 0.2 K at desert rock (DRA) to 2.6 K at BND. Site-specific variations highlight the method's sensitivity to surface heterogeneity and vegetation dynamics. All methods exhibit comparable performance at night (ΔRMSE < 0.15 K).**

***Index Terms*—Land surface temperature (LST), outlier detection, uncertainty, validation.**

Received 8 July 2025; revised 24 September 2025; accepted 20 October 2025. Date of publication 23 October 2025; date of current version 7 November 2025. *(Corresponding author: Zhao-Liang Li.)*

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Digital Object Identifier 10.1109/TGRS.2025.3624786

I. Introduction

LAND surface temperature (LST) is a crucial parameter in the exchange of water and heat between the Earth's surface and the atmosphere [1], [2]. It was used in a wide range of applications, e.g., urban thermal environment analysis [3], [4], drought monitoring [5], [6], and global climate change [7]. LST has been recognized by the Global Climate Observing System (GCOS) as one of the 54 essential climate variables (ECVs) [8].

Multiple research institutions have released various satellite-derived LST products, e.g., Landsat [9], Moderate Resolution Imaging Spectroradiometer (MODIS) [10], [11], [12], and VIIRS [13], [14] LST products. The temperature-based validation approach was widely used to evaluate the accuracy of these LST products against in situ measurements [15], [16], [17], [18], [19]. To address validation accuracy degradation caused by cloud contamination artifacts and satellite–ground spatiotemporal mismatch errors, conventional outlier detection methods have been routinely applied in satellite-derived LST validation using in situ measurements. These primarily include: 1) the three-sigma edit rule [20] based on Gaussian distribution assumptions, i.e., the mean-based method and 2) the three-Hampel identifier [21] employing median absolute deviation, i.e., the median-based method. While the parametric mean-based method provides theoretical simplicity through its mean $\pm$ 3$\times$ standard deviation (SD) criterion, its effectiveness critically depends on normal distribution adherence and becomes vulnerable to outlier induced bias. Conversely, the nonparametric median-based approach demonstrates distributional robustness, particularly advantageous for handling skewed datasets or contaminated samples. However, both conventional outlier detection approaches are based solely on the degree of deviation within statistical data itself, without considering the uncertainties associated with satellite-derived and ground-based LST. This limitation could result in the biased identification of outliers in the validation of satellite-derived LST using in situ measurements.

It is well-established that inherent uncertainties occur in the processes of retrieving LST from satellite data and calculating LST through ground-based measurements. Previous studies have conducted multiple analyses to quantify uncertainties in satellite-derived LST and ground-based observations. However, existing methodologies predominantly addressed

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TABLE I
MODIS PRODUCTS USED IN THIS STUDY

Product name	Available parameters	Spatial resolution	Temporal resolution	Product level
MOD021KM	Radiance	1000 m	5 minutes	L1B
MOD11_L2	VZA and view time	1000 m	5 minutes	L2
MOD03	Latitude and longitude	1000 m	5 minutes	L2
MCD12Q1	Land cover type	500 m	Yearly	L3
MOD13A3	NDVI	1000 m	Monthly	L3

each uncertainty source in isolation. For example, Freitas et al. [22] proposed a per-pixel uncertainty estimation method for SEVIRI-derived LST. Gui et al. [23] established a pixel-level uncertainty quantification method for the generalized split-window (GSW) algorithm. Laraby and Schott [24] developed a protocol for Landsat thermal data uncertainty assessment. Hulley et al. [25] introduced the temperature emissivity uncertainty simulator (TEUSim) for the temperature and emissivity separation algorithm. Similarly, ground-based uncertainty analyses have been conducted through a comprehensive assessment of thermal site variability [26] and systematic estimation of total measurement uncertainty [27]. Despite these methodological advancements, the integration of both uncertainty components in satellite-derived LST validation remains underdeveloped. Ghent et al. [28] compared the SD of satellite–ground LST discrepancies with the combined total uncertainty derived from both measurement sources, demonstrating an effort in uncertainty integration.

This study aims to propose an uncertainty-based outlier detection method for improving the validation accuracy of satellite-derived LST using in situ measurements. The remainder of this article is structured as follows. Section II introduces the data used in this study. In Section III, we present the methodology for the quantification of satellite-derived LST and in situ measurements and the uncertainty-based outlier detection method. The results and discussion are presented in Sections IV and V. The conclusions are drawn in Section VI.

II. DATA

A. MODIS Data

To advance the understanding of global climate dynamics and anthropogenic influences on Earth's systems, NASA's Earth Observing System (EOS) deployed two complementary polar-orbiting satellites: Terra (December 1999) and Aqua (May 2002). These sun-synchronous platforms operate in coordinated morning and afternoon orbits, with Terra's equatorial crossing at 10:30 A.M. local solar time and Aqua's at 1:30 P.M., enabling diurnal environmental monitoring through their one–two-day global coverage cycle.

The MODIS, a cornerstone instrument aboard both satellites, provides multispectral observations through 36 discrete spectral bands spanning visible–thermal infrared wavelengths (0.4–14.5 μm). With a wide 2330-km swath width and selectable spatial resolutions of 250 m, 500 m, and 1 km, MODIS delivers comprehensive datasets for quantifying biophysical parameters across terrestrial, atmospheric, and oceanic domains.

All MODIS data products are archived in HDF-EOS format and publicly accessible through NASA's Earthdata platform (https://search.earthdata.nasa.gov/). Table I summarizes the specific MODIS datasets used in this study. Specifically, the radiance data from the 1-km thermal infrared bands 31 and 32 of MOD021KM were used to calculate the brightness temperature at the top of atmosphere (TOA). The MOD11_L2 product provided the view zenith angle (VZA) and view time data required for LST retrieval and uncertainty estimation. Geolocation information was obtained from MOD03, while land cover types at the in situ sites were determined using MCD12Q1, as described in Section II-D. The normalized vegetation index (NDVI) provided by the MOD13A3 product was used to estimate the land surface emissivity (LSE) of MODIS channels 31 and 32.

B. ASTER GED Product

As a comprehensive emissivity compilation, the Advanced Spaceborne Thermal Emission and Reflection Radiometer Global Emissivity Dataset (ASTER GED) synthesizes global observations from ASTER acquired from 2000 to the present [29]. This dataset serves as a critical input for refining LST derivations. Validation studies confirm ASTER GED's performance in hyper-arid to semiarid environments, establishing it as the preferred reference for unvegetated terrain analysis. However, the dataset's reliance on multitemporal emissivity composites introduces inherent limitations, as temporal averaging effectively eliminates seasonal vegetation dynamics and renders it unsuitable for ecosystems with periodic phenological variations. To address this constraint, our methodology strategically applies ASTER GED exclusively to unvegetated pixels, which are subsequently integrated with ancillary variables to estimate MODIS thermal infrared channels 31 and 32 emissivities. Moreover, since broadband emissivity variability contributes less than 0.3 K to LST error for typical conditions [30], date-specific emissivity values are not required in this study.

C. ERA5 Atmospheric Profile

Atmospheric water vapor (AWV) is a crucial parameter in LST retrieval from satellite thermal infrared (TIR) data. This study used european centre for medium-range weather

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TABLE II
DETAILED INFORMATION ABOUT SEVEN SURFRAD SITES

Site	Latitude	Longitude	Elevation	Land cover type
BND	40.0516° N	88.3733° W	230 m	Croplands
TBL	40.1256° N	105.2378° W	1689 m	Grasslands
DRA	36.6232° N	116.0196° W	1007 m	Open shrublands
FPK	48.3080° N	105.1018° W	634 m	Grasslands
GWN	34.2547° N	89.8729° W	98 m	Cropland/natural vegetation mosaic
PSU	40.7203° N	77.9310° W	376 m	Cropland/natural vegetation mosaic
SXF	43.7343° N	96.6233° W	473 m	Croplands

forecasts (ECMWF) state-of-the-art ERA5 reanalysis product to estimate temporally and spatially collocated AWV measurements synchronized with satellite overpasses. ERA5 achieves enhanced spatiotemporal resolution (0.25° grid and hourly outputs) while preserving continuous global coverage spanning 1950 to the present. Operational implementation involves extracting vertical profiles of geopotential height, air temperature, and relative humidity across 37 pressure levels (1)–1000 hPa) through ECMWF's public Web API. These data were input into MODTRAN 5.2 radiative transfer modeling to estimate AWV.

D. Ground-Based Measurements

Operational since 1993 under NOAA's administration, the surface radiation budget network (SURFRAD) transitioned from triminute to per-minute reporting intervals in 2009 to enhance temporal resolution. This benchmark network delivers comprehensive measurements of key meteorological parameters including broadband incoming/outgoing radiation fluxes, 2-m air temperature, and relative humidity [31]. Its multivariate datasets serve critical roles in satellite product validation, climate model evaluation, hydrological cycle analysis, and terrestrial–atmospheric interaction studies.

The measurement infrastructure employs Eppley precision pyranometers strategically mounted on 10-m instrument towers. These hemispheric radiometers achieve an effective footprint diameter of 30–45 m through optimized sensor geometry, enabling effective data representation within 70 × 70 m grid cells—a spatial resolution particularly valuable for mesoscale model validation [16]. To maintain World Meteorological Organization (WMO) traceability standards, SURFRAD implements rigorous quality assurance protocols. The reference radiometric triad undergoes annual traceable calibrations at the World Radiation Center's Physical Meteorological Observatory (PMOD), ensuring long-term measurement stability better than ±2% across solar spectrum measurements.

Currently comprising seven strategically distributed stations across various ecoregions, the network captures diverse U.S. climate regimes from continental (Bondville, IL, USA) to semiarid (Desert Rock, NV, USA) environments. Table II presents the detailed information about seven SURFRAD sites.

III. METHODOLOGY

A. Uncertainty of MODIS-Derived LST

The GSW algorithm proposed by Wan and Dozier [10] was used to retrieve LST from MODIS TIR data in the bands 31 and 32. The GSW algorithm is written as

$$T_s = A_0 + \left(A_1 + A_2\frac{1-\varepsilon}{\varepsilon} + A_3\frac{\Delta\varepsilon}{\varepsilon^2}\right)\frac{T_{31}+T_{32}}{2} + \left(A_4 + A_5\frac{1-\varepsilon}{\varepsilon} + A_6\frac{\Delta\varepsilon}{\varepsilon^2}\right)\frac{T_{31}-T_{32}}{2} \quad (1)$$

where T_s is the retrieved MODIS LST, T_{31} and T_{32} are the brightness temperatures at the TOA in the MODIS TIR bands 31 and 32, respectively, ε is the average of LSE in the MODIS TIR bands 31 and 32, $\varepsilon = (\varepsilon_{31} + \varepsilon_{32})/2$, $\Delta\varepsilon$ is the difference of LSE in these two bands, $\Delta\varepsilon = \varepsilon_{31} - \varepsilon_{32}$, and A_0–A_6 are the regression coefficients of the GSW algorithm.

Gui et al. [23] proposed a parameterized model for pixel-by-pixel quantifying the uncertainty of MODIS-derived LST using the GSW algorithm. The parameterized model is expressed as (2), shown at the bottom of the next page, where u_{sat} is the uncertainty of MODIS-derived LST, T is the average of brightness temperature at the TOA in the MODIS TIR bands 31 and 32, $T = (T_{31} + T_{32})/2$, ΔT is the difference of brightness temperature at the TOA in these two bands, $\Delta T = T_{31} - T_{32}$, $u_{\varepsilon_{31}}$ and $u_{\varepsilon_{32}}$ are the LSE uncertainties in the MODIS TIR bands 31 and 32, respectively, $u_{T_{31}}$ and $u_{T_{32}}$ are the brightness temperature uncertainties (i.e., NEΔT) in the MODIS TIR bands 31 and 32, respectively, and B_0–B_5 are the regression coefficients of the parameterized model.

The regression coefficients (A_0–A_6) of the GSW algorithm and those (B_0–B_5) of the parameterized model can be determined by a simulation dataset. The simulation dataset was constructed by integrating a comprehensive collection of global atmospheric profiles from the thermodynamic initial guess retrieval 2000 (TIGR2000) database with surface emissivity spectra extracted from the ASTER spectral library [23]. These inputs, representing diverse atmospheric conditions and surface characteristics, were processed through the MODTRAN 5.2 radiative transfer model to generate simulated radiative transfer scenarios. For each prescribed VZA among the six standard configurations (0°, 33.56°, 44.42°, 51.32°, 56.25°, and 60°), the regression coefficients A_0–$A6$ and B_0-B5 were determined through least-squares fitting of the simulation

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dataset. This process accounted for the stratified parameter spaces of both LST and AWV.

The parameter space was segmented with overlapping intervals to ensure continuity in subsequent retrievals. AWV values were categorized into six consecutive subranges with 0.5-g/cm^2 overlaps: 0–1.5, 1.0–2.5, 2.0–3.5, 3.0–4.5, 4.0–5.5, and 5.0–6.5 g/cm^2. Similarly, LST values were divided into five consecutive subranges with 5-K overlaps: <280, 275–295, 290–310, 305–325, and >320 K. This interval-overlapping scheme enhances the stability of LST retrievals across different atmospheric and surface conditions while maintaining physical consistency in the inversion process.

B. Uncertainty of Ground-Based LST

The ground-based LST can be calculated from the upwelling and downwelling longwave radiations using the Stefan–Boltzmann law [19]

$$T_g = \left[\frac{R_u - (1-\varepsilon_b)R_d}{\sigma\varepsilon_b}\right]^{1/4} \tag{3}$$

where T_g is the ground-based LST, R_u and R_d are the upwelling and downwelling longwave radiations, respectively, σ is the Stefan–Boltzmann constant (5.67×10^{-8} W/m^2/K^4), and ε_b is the surface broadband emissivity, which can be estimated using a linear relationship with the LSEs from the ASTER GED product [32]

$$\begin{aligned}\varepsilon_b = {} & 0.197 + 0.025\varepsilon_{10} + 0.057\varepsilon_{11} + 0.237\varepsilon_{12} \\ & + 0.333\varepsilon_{13} + 0.146\varepsilon_{14}\end{aligned} \tag{4}$$

where ε_{10}–ε_{14} are the LSEs of the ASTER bands 10–14.

The main uncertainty sources of ground-based LST primarily arise from the uncertainties in upwelling longwave radiation, downwelling longwave radiation, and surface broadband emissivity. According to the law of error propagation, the uncertainty of ground-based LST is expressed as

$$u_{\text{ground}} = \sqrt{u_{\text{up}}^2 + u_{\text{down}}^2 + u_{\text{ems}}^2} \tag{5}$$

where u_{ground} is the uncertainty of ground-based LST, u_{up} and u_{down} are the LST uncertainties due to the uncertainties in upwelling and downwelling longwave radiations, respectively, and u_{ems} is the LST uncertainty associated with surface broadband emissivity uncertainty.

u_{up} and u_{down} are expressed as

$$u_{\text{up}} = \frac{\partial T_g}{\partial R_u}u_{R_u} = \frac{T_g}{4\left[R_u - (1-\varepsilon_b)R_d\right]}u_{R_u} \tag{6}$$

$$u_{\text{down}} = \frac{\partial T_g}{\partial R_d}u_{R_d} = \frac{T_g(\varepsilon_b - 1)}{4\left[R_u - (1-\varepsilon_b)R_d\right]}u_{R_d} \tag{7}$$

where u_{R_u} and u_{R_d} are the uncertainties of upwelling and downwelling longwave radiations, respectively, which are approximately 5 W/m^2 [33].

u_{ems} is written as

$$u_{\text{ems}} = \frac{\partial T_g}{\partial \varepsilon_b}u_{\varepsilon_b} = \frac{T_g(R_d - R_u)}{4\varepsilon_b\left[R_u - (1-\varepsilon_b)R_d\right]}u_{\varepsilon_b} \tag{8}$$

where u_{ε_b} is the uncertainty of surface broadband emissivity, which is the combination of LSE uncertainties in the ASTER bands 10–14 and the uncertainty of the fitting equation [see (4)]. The uncertainty of the ASTER GED product in each band is approximately 0.01 [29], and a root-mean-squared error (RMSE) of 0.005 is given for the fitting equation.

C. Total Uncertainty Associated With the Validation of MODIS-Derived LST Using Ground-Based LST

The uncertainty sources for the validation of MODIS-derived LST using ground-based LST include not only the uncertainties of MODIS-derived and ground-based LST, but also the LST uncertainties associated with temporally and spatially matching MODIS-derived and ground-based LST. Assuming that each uncertainty component is independent mutually, the total LST uncertainty associated with the validation of MODIS-derived LST using ground-based LST is expressed as

$$u_{\text{total}} = \sqrt{u_{\text{sat}}^2 + u_{\text{ground}}^2 + u_{\text{space}}^2 + u_{\text{time}}^2} \tag{9}$$

where u_{total} is the total LST uncertainty, and u_{space} and u_{time} are the LST uncertainties associated with temporally and spatially matching MODIS-derived and ground-based LST, respectively.

For u_{space}, the SD of clear-sky MODIS-derived LST in a window of 3×3 pixels centered at the validation site is estimated [28]. For u_{time}, the SD of ground-based LST within 3 min centered at the MODIS acquisition time is calculated. The choice of a ±3-min window helps avoid significant effects from rapid intraday temperature changes while preserving sufficient matchup samples [34]. As the optimization of the temporal window is not the main focus of this study, we adopted this threshold following previous LST validation practice.

D. Uncertainty-Based Outlier Detection Method

To minimize the impact of possible outliers on the statistics of LST validation, it is necessary to remove outliers due to cloud contamination or spatiotemporal mismatch errors between MODIS-derived and ground-based LSTs.

For each validation matchup, the difference between MODIS-derived and ground-based LSTs is written as

$$\Delta T_i = T_{s,i} - T_{g,i} \tag{10}$$

where ΔT is the difference between MODIS-derived and ground-based LSTs, and i is the ith validation matchup.

$$u_{\text{sat}} = \sqrt{B_0 + \left(B_1\frac{T^2}{\varepsilon^4} + B_2\frac{\Delta T^2}{\varepsilon^4} + B_3\frac{T\cdot\Delta T}{\varepsilon^4}\right)\left(u_{\varepsilon_{31}}^2 + u_{\varepsilon_{32}}^2\right) + \left(B_4\frac{1-\varepsilon}{\varepsilon} + B_5\frac{\Delta\varepsilon}{\varepsilon^2}\right)\left(u_{T_{31}}^2 + u_{T_{32}}^2\right)} \tag{2}$$

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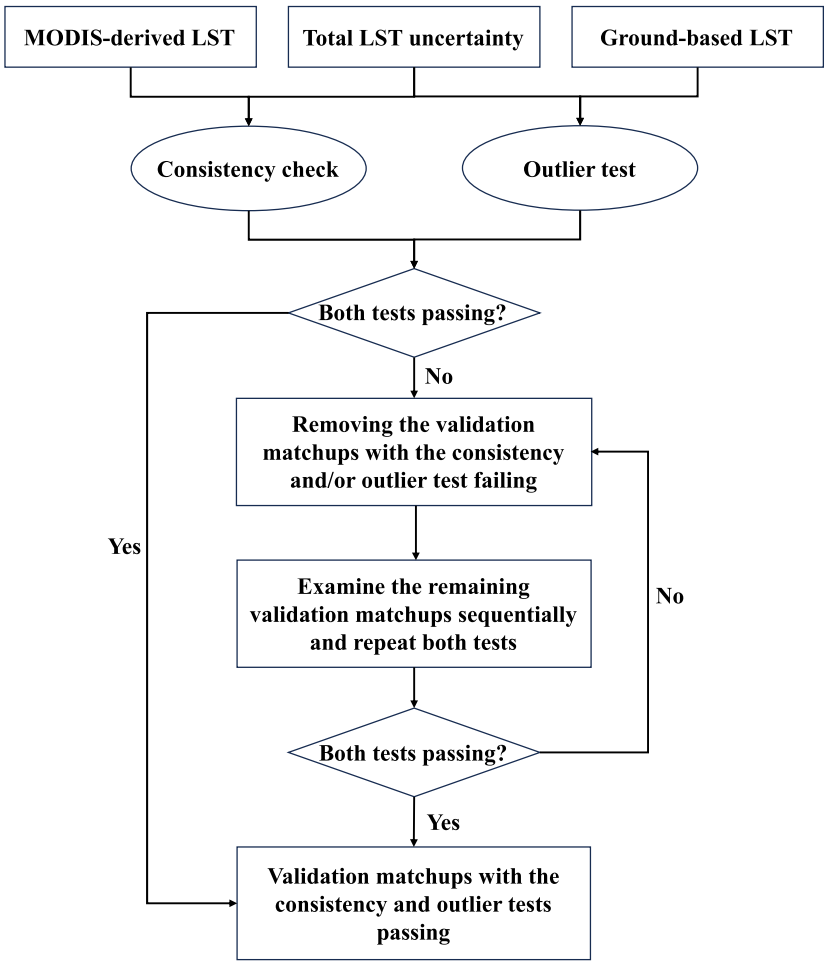


Fig. 1. Flowchart of the uncertainty-based outlier detection method.

Assuming that each validation matchup is independent mutually, the weighted mean of the differences between MODIS-derived and ground-based LSTs is given by

$$\Delta T_{\mathrm{wm}} = \sum_{i=1}^{N} w_i \Delta T_i \tag{11}$$

where ΔT_{wm} is the weighted mean of the differences between MODIS-derived and ground-based LSTs, and N is the number of validation matchup.

The weight w_i of ΔT_i is determined by

$$w_i = \frac{u_{\mathrm{total},i}^{-2}}{\sum_{i=1}^{N} u_{\mathrm{total},i}^{-2}}. \tag{12}$$

The uncertainty $u_{\Delta T_{\mathrm{wm}}}$ associated with the weighted mean ΔT_{wm} is given by

$$u_{\Delta T_{\mathrm{wm}}} = \frac{1}{\sqrt{\sum_{i=1}^{N} u_{\mathrm{total},i}^{-2}}}. \tag{13}$$

A chi-squared test is used to check the consistency of the data sequence $\{\Delta T_i\}$. The observed chi-squared value χ^2_{obs} for the data sequence $\{\Delta T_i\}$ is calculated as

$$\chi^2_{\mathrm{obs}} = \sum_{i=1}^{N} \frac{(\Delta T_i - \Delta T_{\mathrm{wm}})^2}{u_{\mathrm{total},i}^2}. \tag{14}$$

At a confidence level of 0.05, the consistency check fails if the following criterion is fulfilled:

$$probability\left\{\chi^2(v) > \chi^2_{\mathrm{obs}}\right\} < 0.05 \tag{15}$$

where v is the degrees of freedom ($v = N - 1$), and $\chi^2(v)$ is the reference chi-squared value obtained from a lookup table at a confidence level of 0.05 with the degrees of freedom v.

The deviation d_i between ΔT_i and ΔT_{wm} is used to characterize the difference of each validation matchup

$$d_i = \Delta T_i - \Delta T_{\mathrm{wm}}. \tag{16}$$

The uncertainty of d_i is calculated as

$$u_{d,i} = \sqrt{u^2_{\mathrm{total},i} - u^2_{\Delta T_{\mathrm{wm}}}} \tag{17}$$

where $u_{d,i}$ is the uncertainty of d_i.

An outlier is detected according to the following criterion [35]:

$$\frac{|d_i|}{2u_{d,i}} > 1. \tag{18}$$

The main procedures of the uncertainty-based outlier detection method are summarized as follows (Fig. 1):

1) obtaining the difference (ΔT_i) between MODIS-derived ($T_{s,i}$) and ground-based LSTs ($T_{g,i}$) for the ith validation matchup using (10);
2) calculating the total LST uncertainty ($u_{\mathrm{total,i}}$) for the ith validation matchup using (9);

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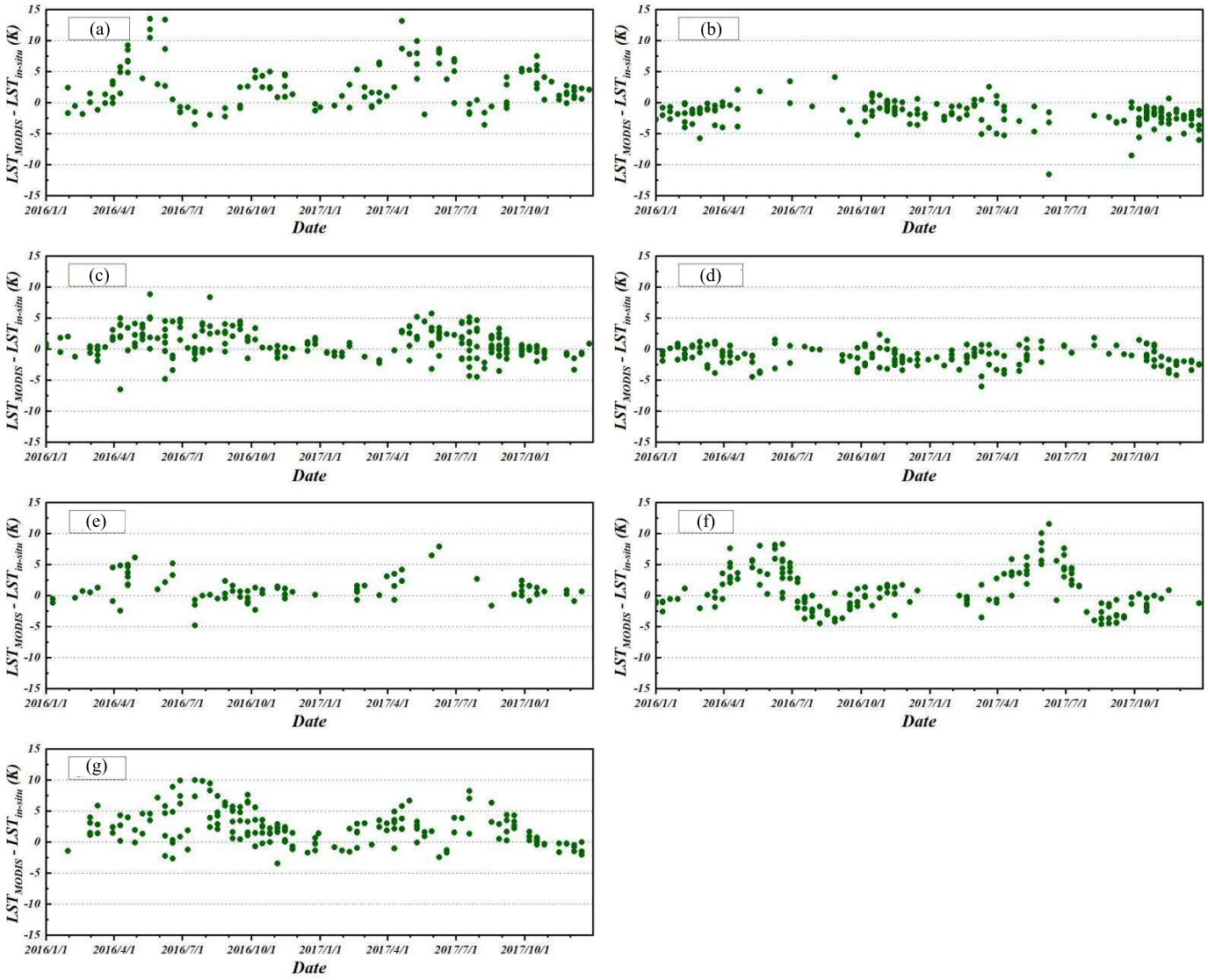


Fig. 2. Temporal dynamics of the differences between MODIS-derived LST before outlier removal and in situ LST during daytime in 2016 and 2017 at the seven SURFRAD sites: (a) BND; (b) DRA; (c) FPK; (d) GWN; (e) PSU; (f) SXF; and (g) TBL.

3) estimating the weight (w_i) of the LST difference (ΔT_i) for the ith validation matchup using (12);
4) calculating the weighted mean (ΔT_{wm}) of the differences between MODIS-derived and ground-based LSTs for all validation matchups using (11), and the uncertainty ($u_{\Delta T_{\mathrm{wm}}}$) associated with the weighted mean using (13);
5) checking the consistency test using (14) and (15), and the outlier test using (16)–(18);
6) removing the validation matchup with the consistency and/or outlier test failing;
7) repeating steps (5) and (6) until all validation matchups pass both tests.

For comparison, two methods widely used to detect outliers were introduced in this study, i.e., the mean- and median-based outlier detection methods. The mean-based outlier detection method detects outliers that deviate by more than three SDs from the mean [20]. The mean and SD are written as

$$\Delta T_{\mathrm{mean}} = \frac{1}{N}\sum_{i=1}^{N}\Delta T_i \tag{19}$$

$$\Delta T_{\mathrm{SD}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\Delta T_i - \Delta T_{\mathrm{mean}})^2} \tag{20}$$

where ΔT_{mean} and ΔT_{SD} are the mean and SD of the data sequence $\{\Delta T_i\}$, respectively.

The median-based outlier detection method rejects all validation matchups outside of three robust SDs (RSDs) from the median [21]. The median and RSD are written as

$$\Delta T_{\mathrm{med}} = median\,(\Delta T_i) \tag{21}$$

$$\Delta T_{\mathrm{RSD}} = 1.4826 \times median\,\{|\Delta T_i - \Delta T_{\mathrm{med}}|\} \tag{22}$$

where ΔT_{med} and ΔT_{RSD} are the median and RSD of the data sequence $\{\Delta T_i\}$, respectively.

IV. Results

A. Temporal Dynamics of LST Differences and Associated Uncertainty

Fig. 2 shows the temporal variations of the discrepancies between MODIS-derived LST before outlier removal and in

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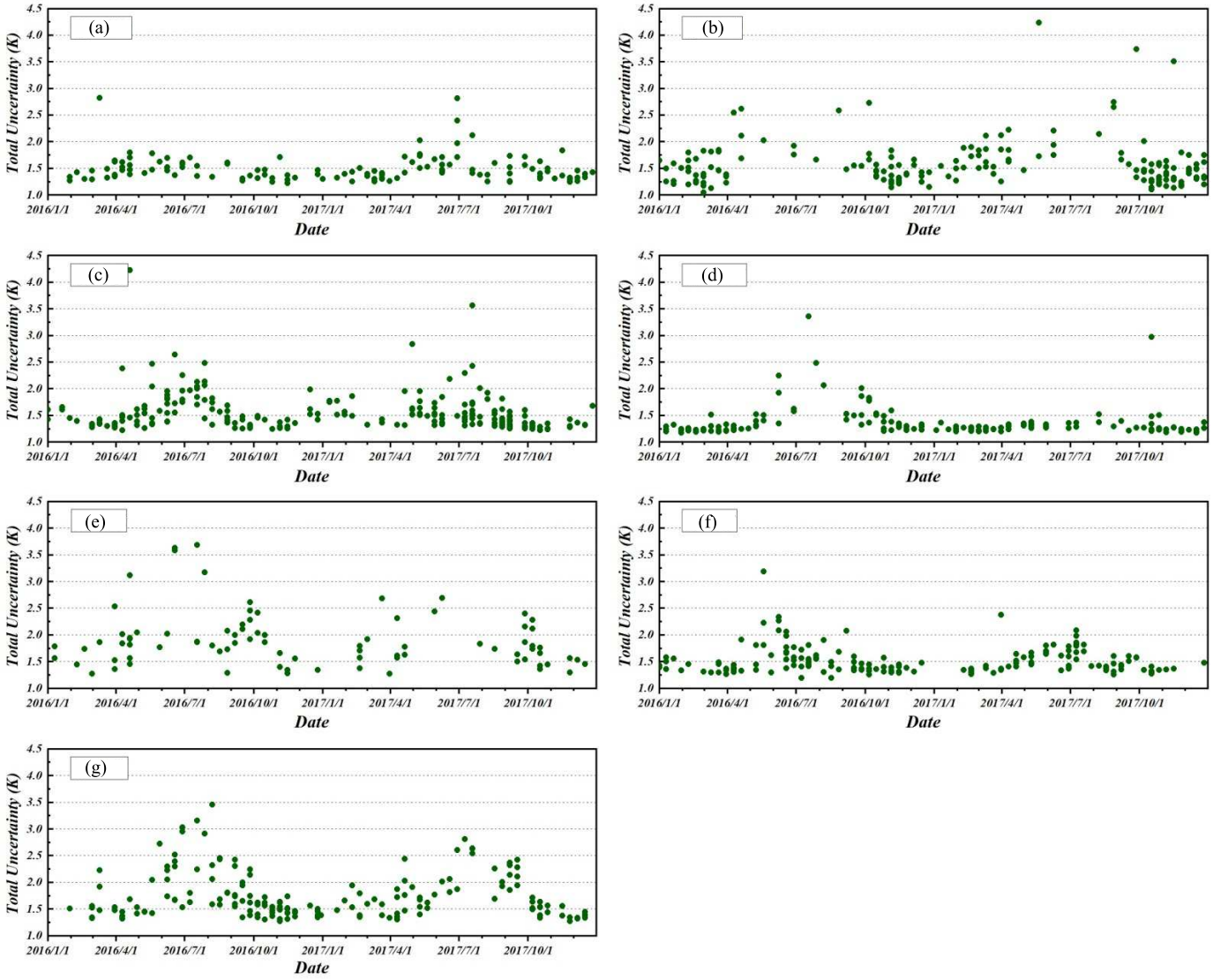


Fig. 3. Temporal dynamics of total LST uncertainty during daytime in 2016 and 2017 at the seven SURFRAD sites: (a) BND; (b) DRA; (c) FPK; (d) GWN; (e) PSU; (f) SXF; and (g) TBL.

situ LST during daytime at SURFRAD sites from 2016 to 2017. Both years exhibit comparable annual patterns in LST differences across the study sites.

At the Bondville (BND) site, pronounced positive biases emerge during May and October, coinciding with postharvest periods when satellite observations primarily detect bare soil surfaces. The observed LST elevation relative to ground measurements stems from the thermal contrast between exposed soil and the vegetated surfaces monitored by site radiometers. These bias decreases to minimal levels during summer crop maturity (July–August) and winter snow cover periods (December–January), when surface characteristics exhibit better sensor-site congruence.

The desert rock (DRA) site demonstrates consistent negative biases throughout most annual cycles, with notable data scarcity during summer months due to persistent cumulus cloud cover. Contrastingly, the Fort Peck (FPK) site exhibits anomalous summer positive biases potentially attributable to differential vegetation patterns caused by bison grazing regimes, creating pronounced thermal contrasts between grazed and ungrazed areas. Winter observations at FPK are limited by frequent stratus cloud formations.

The Goodwin Creek (GWN) site shows minimal seasonal LST variation but experiences summer data gaps correlating with enhanced atmospheric moisture from monsoon systems. The Pennsylvania State University (PSU) site displays elevated positive biases during the agricultural planting season (May–July), potentially linked to surface moisture variations in tilled soils. Summer observations here are constrained by persistent AWV.

The Sioux Falls (SXF) site exhibits marked seasonal cyclicity with May–June positive anomalies transitioning to August–September negative deviations, likely reflecting phenological changes in mixed agricultural systems. Winter data scarcity aligns with frequent frontal cloud systems. The Table Mountain (TBL) site demonstrates heterogeneous summer

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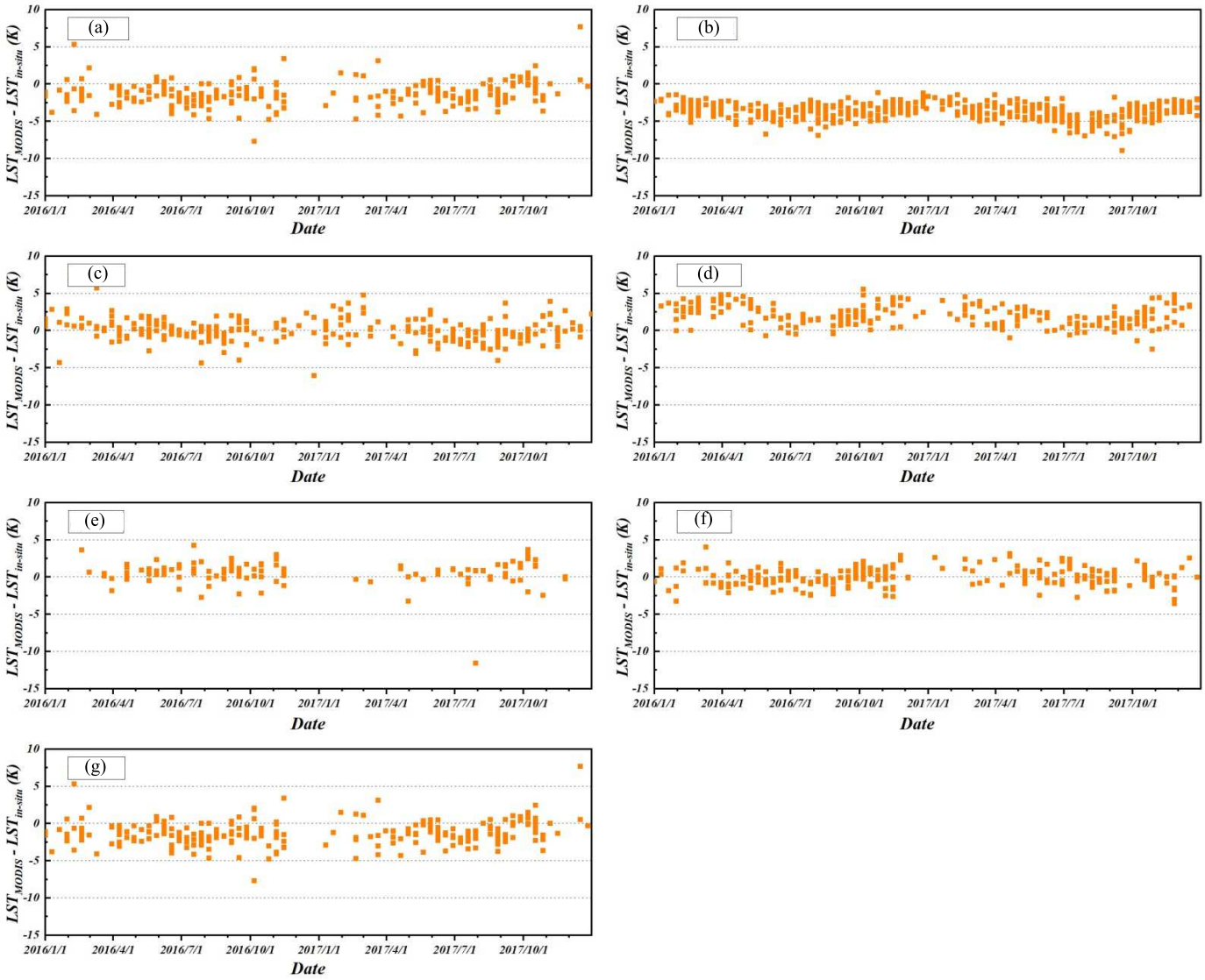


Fig. 4. Temporal dynamics of the differences between MODIS-derived LST before outlier removal and in situ LST at night in 2016 and 2017 at the seven SURFRAD sites: (a) BND; (b) DRA; (c) FPK; (d) GWN; (e) PSU; (f) SXF; and (g) TBL.

biases with winter stability, suggesting potential microclimate influences from local topography.

Fig. 3 presents the temporal evolution of total LST uncertainty during daytime across seven SURFRAD sites during 2016–2017. The total LST uncertainty of most data points at all sites is located between 1 and 2 K. A larger total LST uncertainty can be observed during summer. It could be attributed to the fact that higher LST in summer results in a larger spatial variability effect.

Importantly, the seasonal patterns of uncertainty in Fig. 3 are generally consistent with the bias dynamics observed in Fig. 2. For instance, periods with elevated positive or negative discrepancies (e.g., summer months at BND, FPK, and PSU) coincide with higher total uncertainties, reflecting enhanced spatial heterogeneity of surface temperature and stronger AWV effects. In contrast, during winter months when both LST differences and uncertainties are relatively small, the consistency between satellite and ground observations improves. This correspondence highlights that the quantified uncertainties provide a reasonable explanation for the observed LST deviations.

Fig. 4 shows temporal variations of the differences between MODIS-derived LST before outlier removal and in situ LST at night in 2016 and 2017 at the seven SURFRAD sites. Compared with Fig. 2, nighttime LST differences exhibit no pronounced seasonal variations. With the exception of DRA, GWN, and TBL sites, LST differences at the other sites are distributed around the 0-K baseline without obvious positive/negative biases. This phenomenon may be attributed to reduced spatial variability impacts during nighttime compared to daytime conditions, as structural shading effects, evaporative cooling processes, and surface–air thermal gradients are substantially diminished after sunset.

Three sites exhibit distinct deviation patterns. The DRA station manifests consistent negative biases throughout nighttime periods, aligning with systematic discrepancies documented

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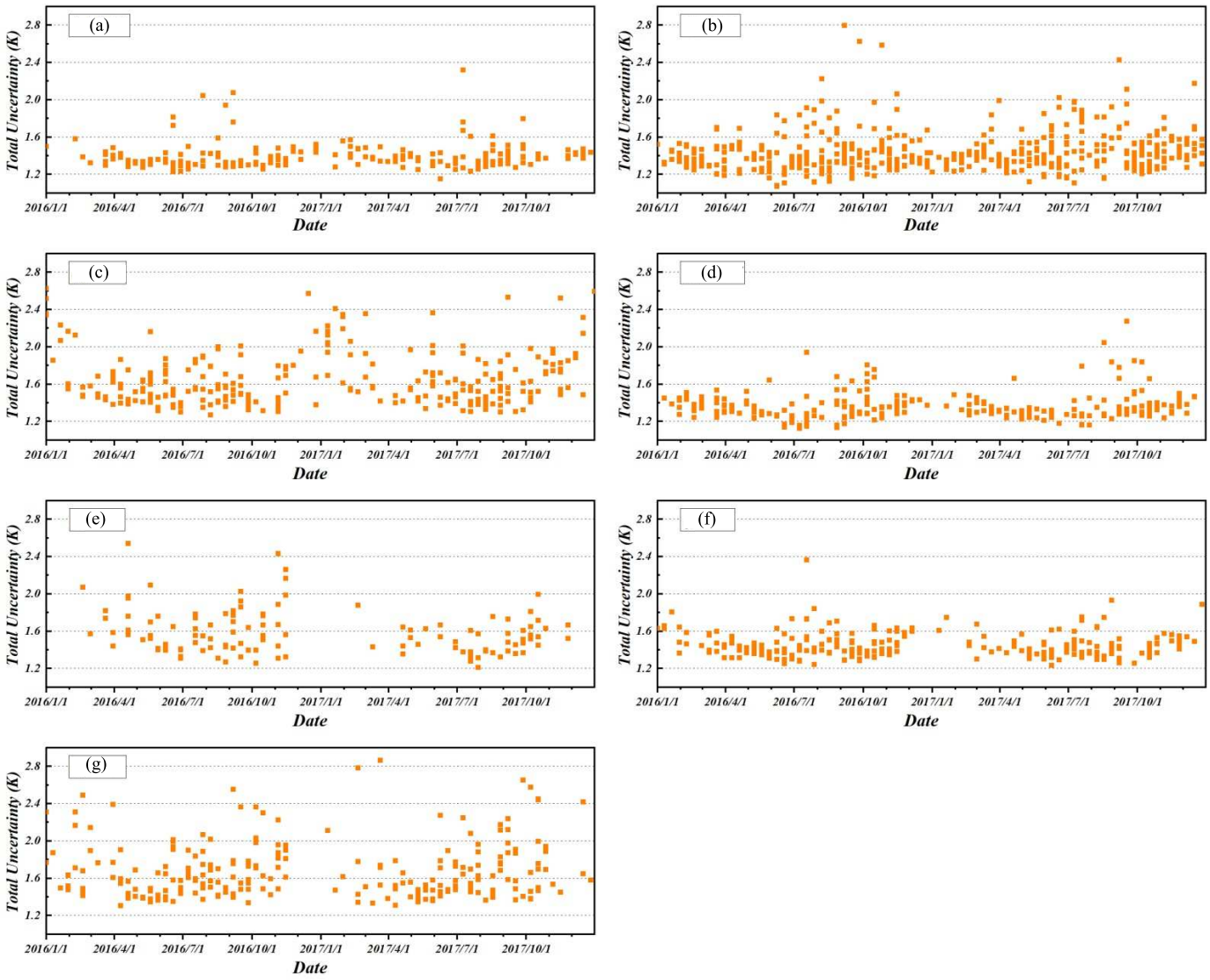


Fig. 5. Temporal dynamics of total LST uncertainty at night in 2016 and 2017 at the seven SURFRAD sites: (a) BND; (b) DRA; (c) FPK; (d) GWN; (e) PSU; (f) SXF; and (g) TBL.

in previous investigations [16], [18], [19]. This discrepancy could potentially stem from observational geometry differences because the hemispherical configuration of ground sensors may observe more bush-covered surfaces compared to the directional measurements obtained by satellite sensors. Conversely, GWN displays persistent positive biases at night, likely resulting from thermal contrast between mixed vegetation types. It could be attributed to the fact that satellite-observed woodland–grassland mosaics maintain higher nighttime temperatures than homogeneous grassland surfaces monitored by ground sensors, as tree canopies typically exhibit thermal inertia closer to ambient air temperatures while short grass undergoes enhanced radiative cooling. The TBL site's negative biases may originate from its topographic position adjacent to a steep plateau edge, where local microclimatic conditions promote cooler thermal emissions detectable by satellite observations. In comparison with the daytime cases, these nighttime deviations are generally smaller and more stable, which can be attributed to the reduced influence of spatial variability after sunset. Structural shading effects, evaporative cooling, and surface–air thermal gradients are all diminished at night, leading to smaller bias magnitudes.

Fig. 5 shows the temporal dynamics of total LST uncertainty during nighttime in 2016 and 2017 at the seven SURFRAD sites. Compared with Fig. 3, the total LST uncertainty of most data points at all sites is less than 1.5 K. This can be explained by the reduced role of spatial variability during nighttime relative to daytime. Importantly, this reduction in uncertainty is consistent with the smaller and more stable nighttime LST differences observed in Fig. 4, supporting the consistency of the uncertainty framework across diurnal conditions.

B. Validation of MODIS-Derived LST With and Without Uncertainty-Based Outlier Removal

Fig. 6 displays the scatterplots of MODIS-derived LST without outlier removal versus in situ LST during daytime in 2016 and 2017 at the seven SURFRAD sites. The best

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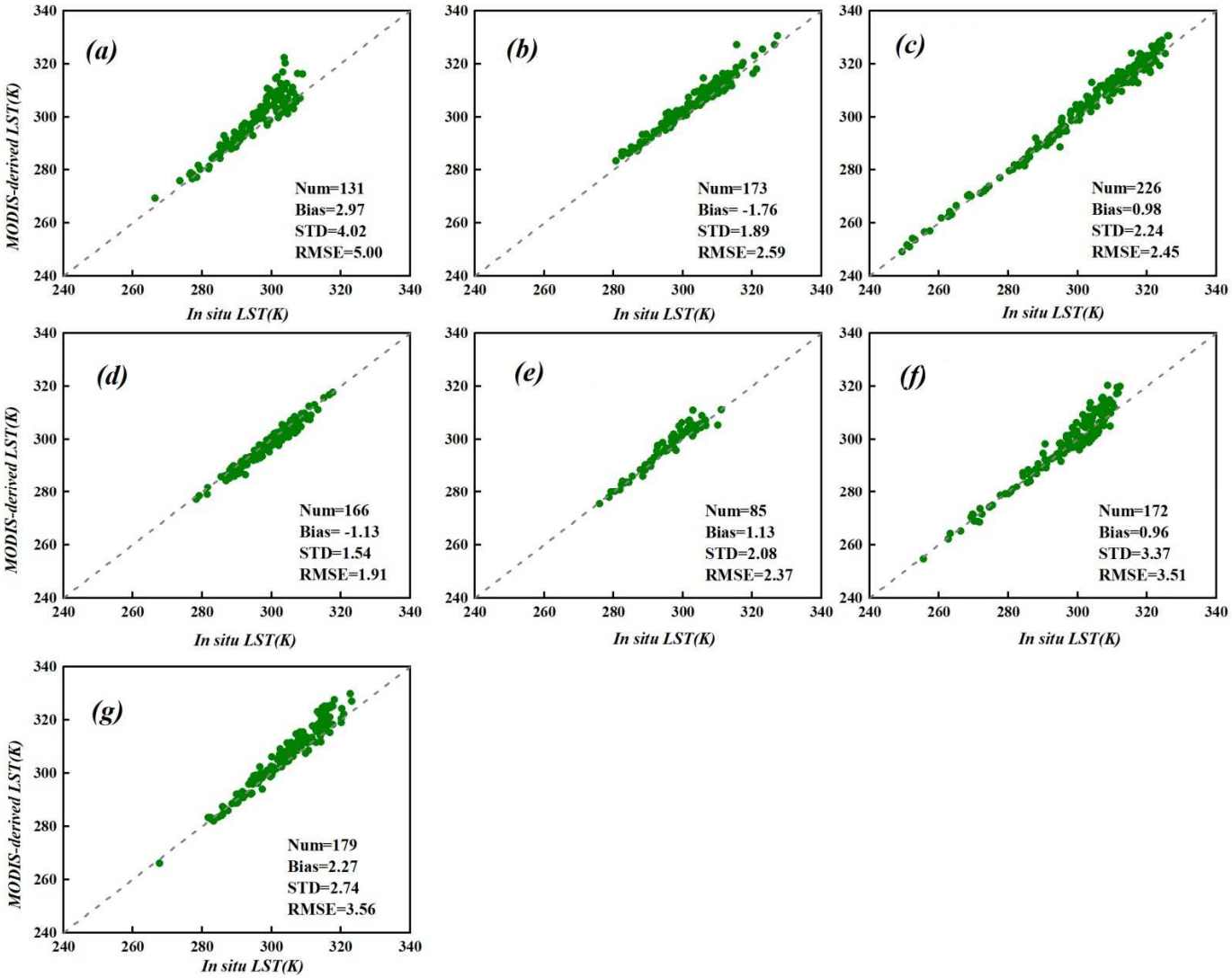


Fig. 6. Scatterplots of MODIS-derived LST without outlier removal versus in situ LST during daytime in 2016 and 2017 at the seven SURFRAD sites: (a) BND; (b) DRA; (c) FPK; (d) GWN; (e) PSU; (f) SXF; and (g) TBL.

validation results are obtained for the GWN site, with bias, standard deviation (STD), and RMSE of approximately −1.1, 1.5, and 1.9 K, respectively, whereas the worst validation results are achieved for the BND site, with bias, STD, and RMSE of approximately 3.0, 4.0, and 5.0 K, respectively. The RMSE values at the other sites are located between 2.4 and 3.6 K.

Fig. 7 displays the scatterplots of MODIS-derived LST with uncertainty-based outlier removal versus in situ LST during the day in 2016 and 2017 at the seven SURFRAD sites. Compared with Fig. 6, it is obvious that the LST validation results with uncertainty-based outlier removal are better than those without outlier removal, especially at the BND and SXF sites. The RMSE at the BND site reduces from approximately 5.0 K without outlier removal to 1.9 K with uncertainty-based outlier removal, and that at the SXF site decreases from approximately 3.5 K without outlier removal to 1.6 K with uncertainty-based outlier removal. The results demonstrate that the uncertainty-based outlier removal method can effectively remove LST outliers.

Fig. 8 shows the scatterplots of MODIS-derived LST without outlier removal versus in situ LST during nighttime in 2016 and 2017 at the seven SURFRAD sites. The best validation results are achieved for the SXF site, with the bias, STD, and RMSE of approximately 0, 1.3, and 1.3 K, respectively, whereas the worst validation results are obtained for the DRA site, with the bias, STD, and RMSE of approximately −3.7, 1.1, and 3.9 K, respectively. Compared with Figs. 6 and 8, except for the DRA and GWN sites, the validation results at night are better than those during the day at the other sites. It could be attributed to the fact that in situ LST measurements during nighttime are more representative of LST at the satellite pixel scale. The DRA site has a large absolute bias of approximately 3.7 K. Similar systematic biases were also observed in previous studies [16], [18], [19]. The reason could be due to the hemispherical in situ sensor observing more bushes than the directional satellite sensor.

Fig. 9 shows the scatterplots of MODIS-derived LST with uncertainty-based outlier removal versus in situ LST at night in 2016 and 2017 at the seven SURFRAD sites. Compared

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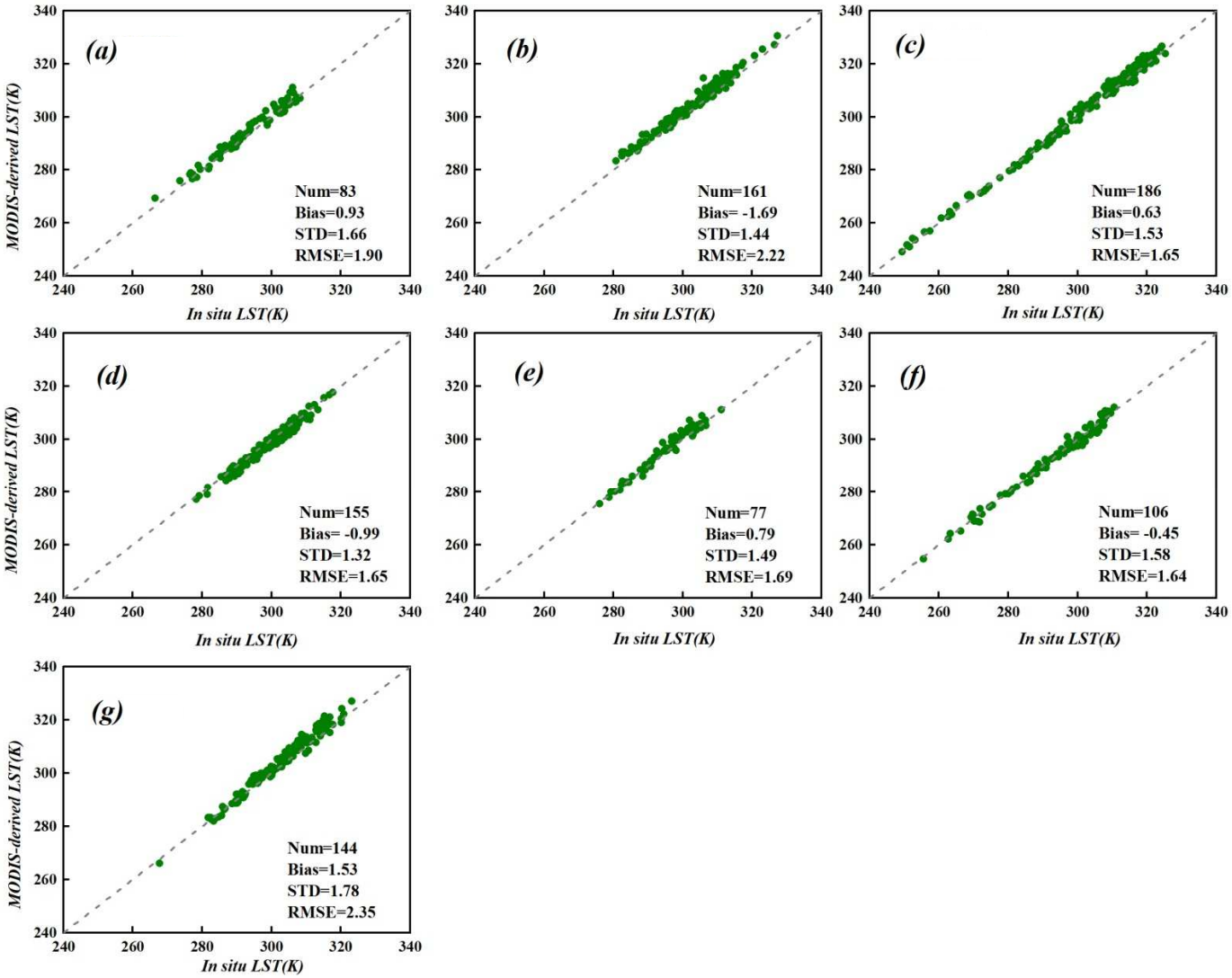


Fig. 7. Scatterplots of MODIS-derived LST with uncertainty-based outlier removal versus in situ LST during daytime in 2016 and 2017 at the seven SURFRAD sites: (a) BND; (b) DRA; (c) FPK; (d) GWN; (e) PSU; (f) SXF; and (g) TBL.

TABLE III

BIAS AND RMSE BETWEEN MODIS-DERIVED LST WITH/WITHOUT UNCERTAINTY-BASED OUTLIER REMOVAL AND IN SITU LST DURING DAYTIME AND NIGHTTIME IN 2016 AND 2017 AT ALL SURFRAD SITES

Day/night	Without outlier removal			With outlier removal		
	Num	Bias	RMSE	Num	Bias	RMSE
Daytime	1132	-0.70 K	3.15 K	912	-0.01 K	1.91 K
Nighttime	1807	0.79 K	2.50 K	1773	0.76 K	2.41 K

with the validation results during daytime, there are only small discrepancies between the LST validation results with and without uncertainty-based outlier removal during nighttime. The change of the RMSE is less than 0.5 K at all sites. Large absolute biases are still observed for the validation results with uncertainty-based outlier removal at the DRA and GWN sites. The results indicate that the uncertainty-based outlier removal method does not simply remove the data points with large biases. Instead, the method evaluates each deviation relative to its propagated uncertainty, and only discards those observations whose standardized residuals are statistically inconsistent with the expected uncertainty distribution. This mechanism ensures that data points with large but explainable biases are retained, while those with implausible deviations are eliminated.

Table III presents the bias and RMSE between MODIS-derived LST with/without uncertainty-based outlier removal and in situ LST during daytime and nighttime in 2016 and 2017 at all SURFRAD sites. During the day, the number of data points was reduced from 1132 without outlier removal to 912 with uncertainty-based outlier removal, and the RMSE was decreased from approximately 3.2 K with-

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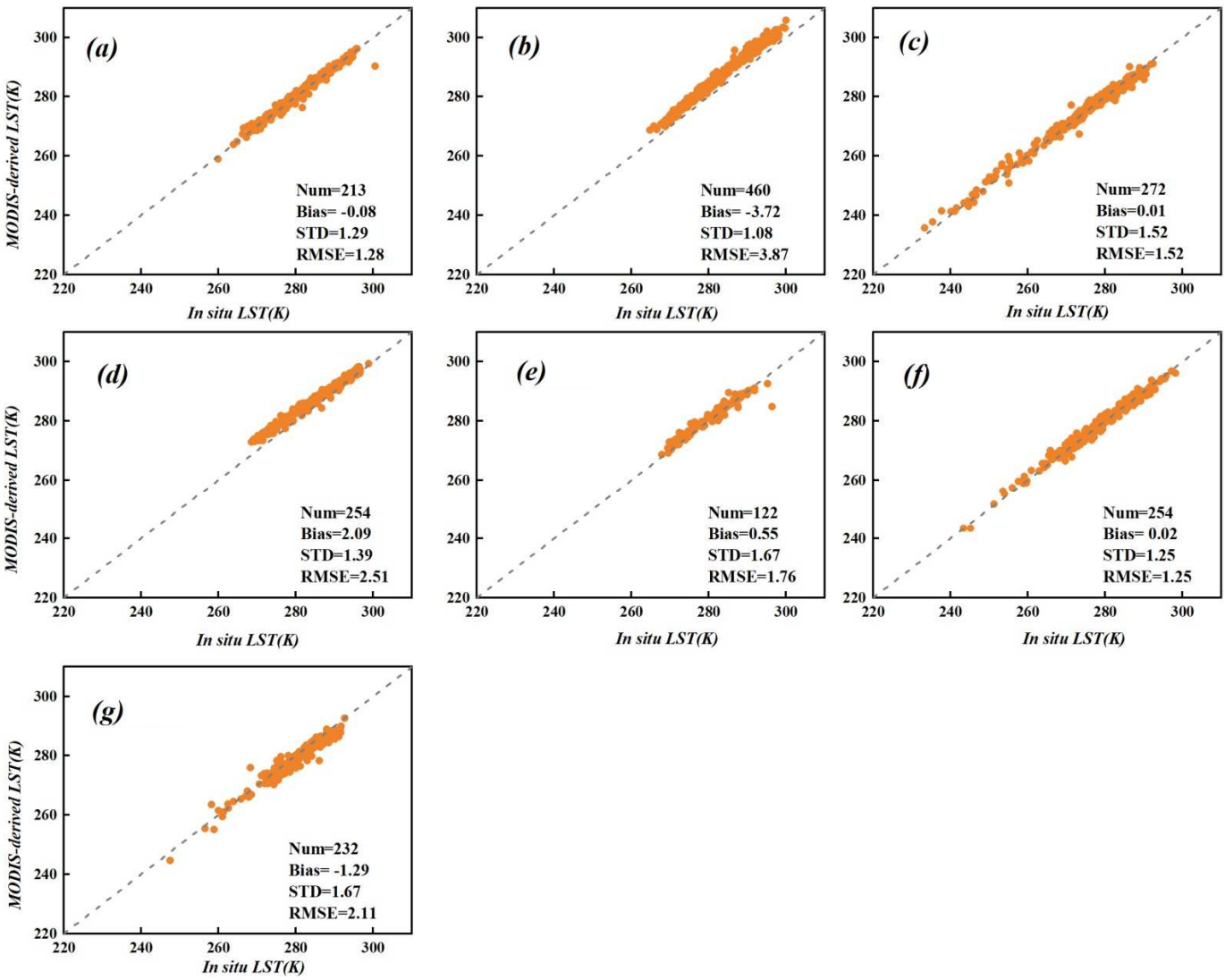


Fig. 8. Scatterplots of MODIS-derived LST without outlier removal versus in situ LST during nighttime in 2016 and 2017 at the seven SURFRAD sites: (a) BND; (b) DRA; (c) FPK; (d) GWN; (e) PSU; (f) SXF; and (g) TBL.

out outlier removal to 1.9 K with uncertainty-based outlier removal. Nevertheless, the change of data points was only 34, and the change of the RMSE was less than 0.1 K at night.

C. Comparison of Different LST Outlier Detection Methods

The performance of the uncertainty-based outlier detection method is further compared with that of the mean- and median-based outlier detection methods [20], [21]. The results are shown in Fig. 10. The absolute bias, STD, and RMSE of the uncertainty-based method are smaller than those of the mean- and median-based methods during daytime, especially at the BND site. The absolute bias, STD, and RMSE at the BND site reduce from approximately 2.7, 3.6, and 4.5 K for the mean- and median-based methods to approximately 0.9, 1.7, and 1.9 K for the uncertainty-based method, respectively. In general, there are no or only small discrepancies in the bias, STD, and RMSE between the mean- and median-based methods during daytime at all sites. Furthermore, there are no significant discrepancies in the bias, STD, and RMSE among the three outlier removal methods during nighttime at all sites.

We further took the LST validation results at the BND and PSU sites as examples to compare the performance of the mean-, median-, and uncertainty-based outlier detection methods. Fig. 11 displays the scatterplots of MODIS-derived LST without outlier removal and with three outlier removal versus in situ LST during daytime at the BND site. The number of data points reduces from 131 without outlier removal to 129 with the mean- and median-based outlier removal and 83 with the uncertainty-based outlier removal. Most outliers are identified by the uncertainty-based method, whereas only two outliers with the largest differences between MODIS-derived LST and in situ LST are detected by the mean- and median-based methods. Furthermore, the RMSE decreases from 5.0 K without outlier removal to 4.5 K with the mean- and median-based outlier removal and 1.9 K with the uncertainty-based outlier removal. For this example, the mean- and median-based methods improve the accuracy of the LST

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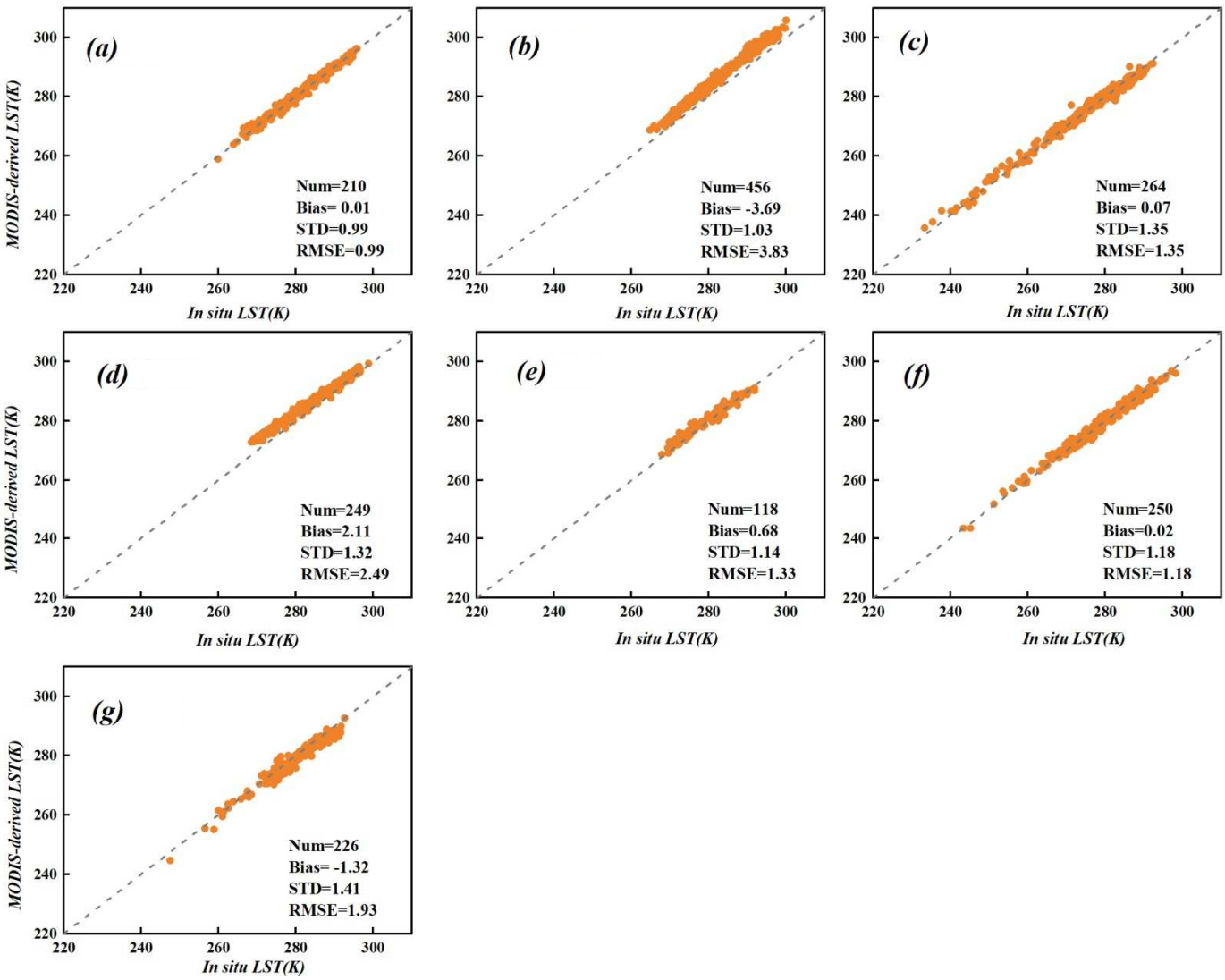


Fig. 9. Scatterplots of MODIS-derived LST with uncertainty-based outlier removal versus in situ LST during nighttime in 2016 and 2017 at the seven SURFRAD sites: (a) BND; (b) DRA; (c) FPK; (d) GWN; (e) PSU; (f) SXF; and (g) TBL.

validation results to a certain extent, but the uncertainty-based method obviously outperforms the mean- and median-based methods.

Fig. 12 shows the scatterplots of MODIS-derived LST without outlier removal and with three outlier removal versus in situ LST during daytime at the PSU site. The number of data points reduces from 85 without outlier removal to 84 with the mean-based outlier removal, to 79 with the median-based outlier removal, and to 77 with the uncertainty-based outlier removal. In addition, the RMSE decreases from approximately 2.4 K without outlier removal to 2.2 K with the mean-based outlier removal, to 1.7 K with the median-based outlier removal, and to 1.7 K with the uncertainty-based outlier removal. For this example, each outlier detection method has a different performance. The accuracies of the LST validation results with three outlier removal are better than those without outlier removal. Furthermore, the uncertainty-based method outperforms the mean-based method, but slightly underperforms the median-based method.

V. Discussion

The rigorous validation of satellite-derived LST not only ensures its reliability for diverse environmental applications, but also provides essential feedback for optimizing LST retrieval algorithms. Among existing validation methods, direct comparison of satellite–ground matchups has become a predominant method for assessing the accuracy of satellite-derived LST. However, a fundamental challenge of the temperature direct comparison method is inherent scale discrepancy between point-scale in situ measurements (typically representing areas <1 m^2) and satellite pixel-level observations (spanning hundreds of square meters to kilometers). This scale mismatch introduces significant uncertainties when directly comparing ground-based LST measurements with their satellite-derived counterparts. To address these validation challenges while ensuring statistical robustness, conventional mean- and median-based outlier detection methods have been developed. These methods effectively mitigate contamination from nonrepresentative samples caused by undetected cloud

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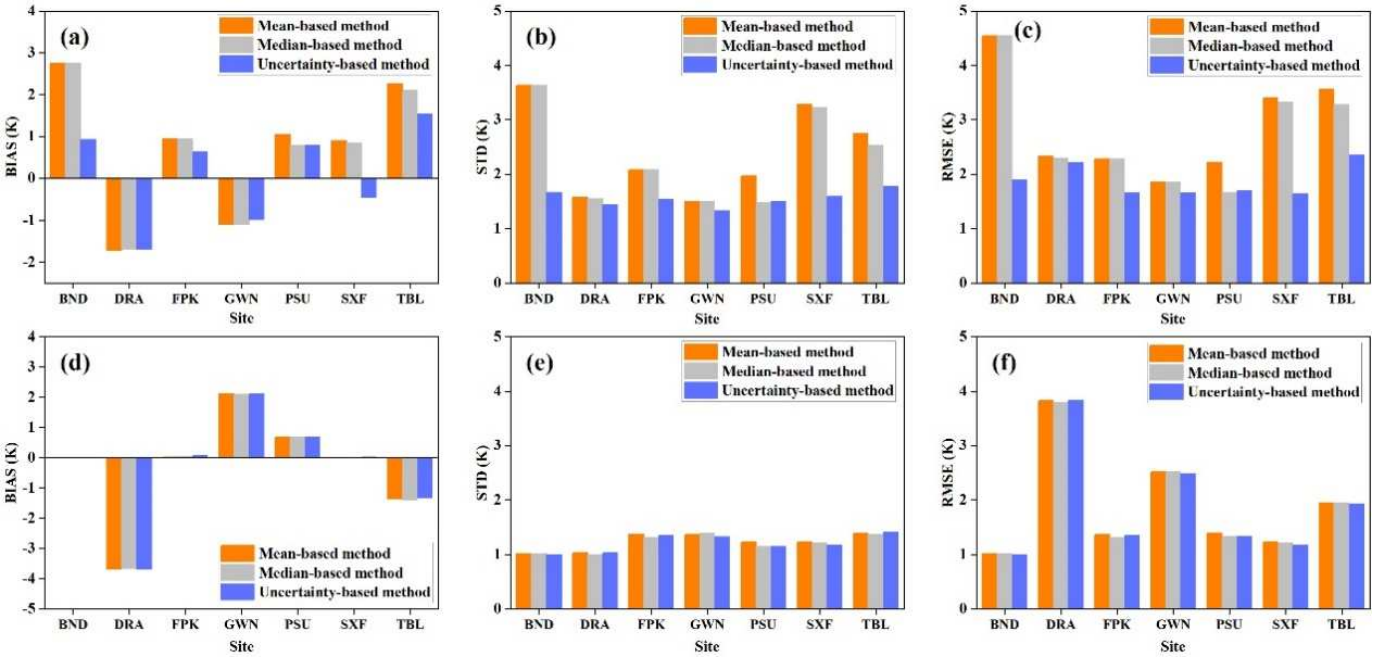


Fig. 10. Bias, STD, and RMSE between MODIS-derived LST with three outlier detection methods and in situ LST during (a)–(c) daytime and (d)–(f) nighttime in 2016 and 2017 at the seven SURFRAD sites.

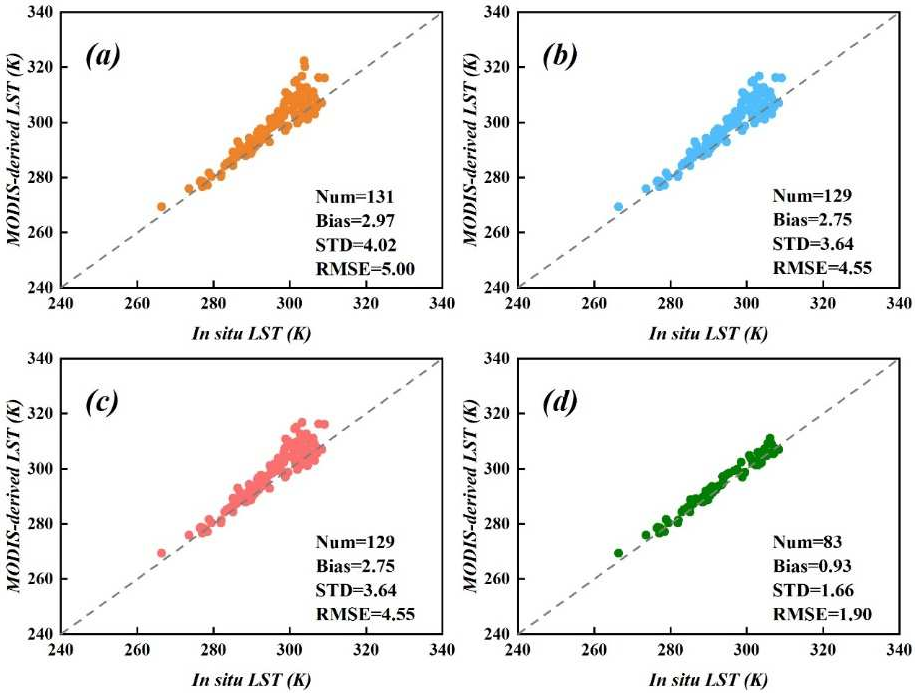


Fig. 11. Scatterplots of MODIS-derived LST versus in situ LST during daytime in 2016 and 2017 at the BND site: (a) without outlier removal; (b) with mean-based outlier removal; (c) with median-based outlier removal; and (d) with uncertainty-based outlier removal.

artifacts, atmospheric disturbances, and spatiotemporal mismatch errors between ground and satellite observations.

The validation of LST products inherently confronts dual uncertainty sources: those arising from satellite retrieval algorithms (including atmospheric correction, sensor noise, and emissivity estimation) and those stemming from ground-based LST computation (involving sensor calibration, emissivity estimation, and surface heterogeneity). This study introduces an uncertainty-based outlier detection method that accounts for both satellite-derived LST uncertainties and ground-measured LST uncertainties. Daytime validation results demonstrate significant accuracy improvement across all SURFRAD sites. Nighttime validation reveals more modest improvements. This diurnal performance directly correlates with the temporal

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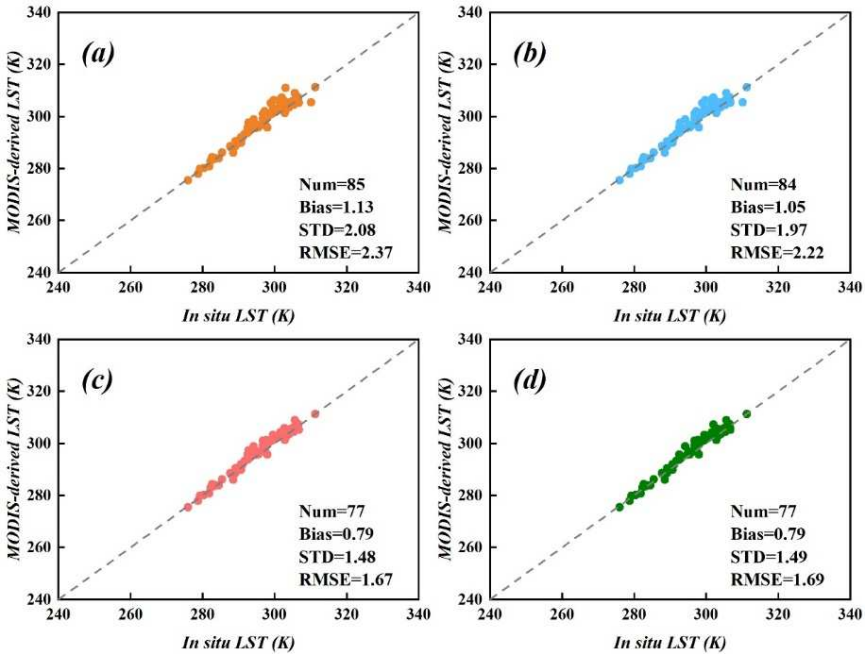


Fig. 12. Scatterplots of MODIS-derived LST versus in situ LST during daytime in 2016 and 2017 at the PSU site: (a) without outlier removal; (b) with mean-based outlier removal; (c) with median-based outlier removal; and (d) with uncertainty-based outlier removal.

patterns of total LST uncertainty budgets. Similar spatially varying validation errors were also reported for new 1-km all-weather LST products [36].

The comprehensive evaluation reveals distinct diurnal patterns in validation efficacy. The uncertainty-based method demonstrates superior validation performance over conventional mean/median-based methods across all SURFRAD sites during daytime, though with notable site-specific variation. The maximum improvement reaches 2.6 K at BND, which is a mid-latitude agricultural site characterized by heterogeneous crop patterns, while minimal enhancement (0.2 K) occurs at DRA with naturally homogeneous arid terrain. This geographical discrepancy likely stems from differential landscape complexity, where the uncertainty method's advantage becomes pronounced in environments with high subpixel thermal variance. However, nighttime comparisons show methodological equivalence (ΔRMSE <0.15 K between approaches), attributable to thermal homogenization processes.

While demonstrating methodological promise, two critical implementation constraints emerge: 1) algorithm transparency requirement: current operational LST products (e.g., MODIS C6) lack complete uncertainty quantification metadata essential for our method, necessitating customized retrievals using the GSW algorithm with documented error propagation characteristics [37] and 2) validation scope limitation: direct application requires priori knowledge of radiometric uncertainty components, atmospheric correction residuals, and land cover-specific emissivity errors, which are typically unavailable in standard LST products.

In addition to the retrieval, ground, and spatiotemporal matching errors explicitly considered in our budget, other sources may also contribute to LST uncertainty. These include radiometric calibration drift of the sensor, seasonal variations in surface emissivity (especially for vegetated areas), and subpixel mixing of heterogeneous surfaces. While these effects are expected to be minor at the relatively homogeneous SURFRAD sites, they may become significant in heterogeneous or long-term applications. Future work should, therefore, incorporate seasonally varying emissivity products, high-resolution data for subpixel variability, and calibration monitoring to achieve a more comprehensive uncertainty assessment. In particular, applying the proposed method to heterogeneous landscapes such as plateaus, forests, or urban areas requires caution, as complex topography, dense canopy structure, and strong subpixel contrasts can amplify these additional uncertainties. Future work should, therefore, integrate seasonally varying emissivity products, high-resolution datasets for subpixel variability, and calibration monitoring to ensure robust application in such heterogeneous environments.

VI. Conclusion

This study proposes an uncertainty-based outlier detection method for satellite-derived LST validation. By integrating both satellite-derived and ground-based LST uncertainty components to estimate total uncertainty budgets, the proposed approach effectively identifies outliers through comprehensive consistency checks and statistical rejection tests. Three principal findings emerge from our investigation.

1) Temporal analysis reveals distinct diurnal patterns in LST discrepancies at different sites. Daytime differences between satellite and ground observations demonstrate phenological sensitivity, with marked positive biases emerging during May and October postharvest periods when sensors predominantly detect bare soil surfaces. These biases diminish to minimal levels during peak

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vegetation phases in summer, aligning with crop maturity stages. In contrast, nighttime LST differences remain stable throughout annual cycles, showing minimal seasonal variation.

2) Validation metrics demonstrate significant error reduction through uncertainty-based outlier removal. Daytime implementation achieves notable RMSE improvements across all SURFRAD sites, particularly at BND with a 3.1-K enhancement (from 5 to 1.9 K). The method shows reduced efficacy during nighttime validation, yielding marginal improvements <0.4 K, potentially attributable to diminished thermal contrast and reduced uncertainty components at night.
3) Comparative performance analysis reveals the uncertainty-based outlier detection method's superiority over conventional mean- and median-based outlier detection methods. Daytime applications show consistent outperformance relative to mean- and median-based outlier removal across all test sites, with site-specific RMSE improvements ranging from 0.2 K at DRA to 2.6 K at BND. Nighttime implementations demonstrate comparable performance across different outlier detection methods with ΔRMSE <0.15 K.

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